

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Introduction to the Program

The Department of Computer Science and Engineering aims to graduate highly competent students with the potential of becoming the leaders in their field and also to impart knowledge which allows them to understand the implications of their work on both to themselves as well to the society as a whole.

ULAB CSE, since its inception, has been offering an undergraduate program in Computer Science and Engineering with a focus on developing skills for ICT industries worldwide. The designed program provides students with the fundamental skills needed to become an accomplished IT specialist. The program also aims at a wider knowledge of the subject, rather than a mere understanding of the existing commercial approaches. The degree is designed for a wide range of students, including those who wish to pursue a career in software engineering, computer programming or related areas, and also those who wish to move on to study for higher degrees in the computing and information technology areas, either at home or abroad.

The department believes in a philosophy of Active Learning which is reflected in our curriculum and teaching styles. The Computer Science and Engineering (CSE) curriculum has been designed with both theoretical and industry-oriented courses. The department provides state of the art laboratory facilities, with guidance from the top-notch faculties, to bring out the best from the students. Departmental active collaboration with local and international reputed companies is keeping both CSE students and faculty updated with industry innovations and requirements. In their last academic year, students will be able to choose a concentration area or any other elective courses to focus their career according to their interest. Furthermore, the Liberal Arts curricular framework will help the students to acquire an openness of vision to become adaptable and ever-effective professionals of this century and for the time to come. The program objectives address our goal of graduating highly competent students with the potential of becoming leaders in their careers and add value to society.

Vision

To produce highly competent graduates who will become leaders in Computer Science and Engineering.

Mission

- To prepare our students to meet high standards of excellence for professional career advancement.
- To create and disseminate new knowledge through basic and applied research in the field of Computer Science and Engineering.
- To build strong relationships between industry and academia, eliminating the gap.

Objectives of the Program

The program objectives, as derived from the departmental mission statement, address our goal of graduating highly competent students with the potential of becoming leaders in the field of their careers and understanding the implications of their work on both to themselves and to society as a whole. The program objectives are:

Foundation: To provide graduates with a solid foundation in the principles and practices of computer science, including mathematics, physical sciences, and basic engineering. This foundation is necessary to succeed in more advanced Computer Science and Engineering coursework where students will apply these principles and practices to a variety of complex problems, and which will further develop their analytical and creative problem-solving skills and enable them to work successfully on multidisciplinary teams, in both classroom and laboratory settings.

Breadth: To provide graduates with sufficient breadth in computer science and related technical disciplines. This breadth is required not only to understand engineering trade-offs that cross disciplines (for example, hardware and software boundaries), but also to participate effectively in multidisciplinary teams. To also provide students with breadth in the liberal arts and social sciences to develop their awareness of societal issues; to develop effective oral and written communication skills; and to develop an understanding of professional and ethical issues related to computing.

Depth: To provide our graduates with advanced training in focused areas of their choice. This depth is necessary to solve complex real-world engineering problems and prepare them to contribute to a specific discipline within computer science and possibly pursue advanced study or research.

Degree Requirements for Bachelor of Science in CSE

To obtain the BSc degree in CSE, students will have to successfully complete minimum 140 credits. The credit and course requirement for the CSE degree program is shown in the table below.

Categories	Credits	Remarks
General Education Courses (GED)	24	5 Major + 3 Elective
Major Core Courses	67	Includes 4 credit Capstone Project, 1 credit Internship
Essential Skills (ESK)	0	Non-Credit
Major Elective Courses	13	4 Theory + 1 Lab
Other Engineering	8	2 Theory + 2 Lab from EEE
Optional/Minor*	9	3 Theory
Basic Science	7	2 Theory + 1 Lab
Mathematics	12	4 Theory
TOTAL	140	

*Students willing to minor must take a total of 5 courses (15 credits).

General Education (GED) Courses

Course Category	No of Courses	Course Code	UNESCO Code	Course Title	Credits
Pre-Uni (based on admission test performance)	1	ELL0099	0231-000-0099	Remidial English	Non Credit
General Education Foundation Courses (GEF)	2	GEF1101	0231-000-1101	Academic English I	2X3=6
		GEF1201	0231-000-1201	Academic English II	
Essential Skills	4	ESK 1110	0031-000-1110	Study Skills	Non Credit
		ESK1111	0031-000-1111	Healthy Life Skills	
		ESK1112	0031-000-1112	Social Skills	
		ESK1113	0031-000-1113	Professional Skills	

ULAB Core Courses	3	UCC1101	0232-000-1101	Bangla Bhasha O Sahitya	3X3=9
		UCC1201	0222-000-1201	History of the Emergence of Independent Bangladesh	
		GED 2159	0223-000-2159	Professional Ethics	
General Education (GED) Elective/ Optional Courses (GED) one from each tier (Tier I Tier II and Tier III)	3	Tier I	Course-wise (varies)	Arts and Humanities	3x3 = 9
		Tier II	Course-wise (varies)	Social Science	
		Tier III	Course-wise (varies)	Natural Science	

Note: Placement in Pre-Uni course, ELL 099: Remedial English, will be based on the admission test.

Note: ** GEF 2159 for students from CSE department

GED Electives (3 courses / 9 credits)

GED Electives course offerings will vary from term to term. Students will choose a Arts and Humanities course from a list of courses and the other two courses, GED 2243 and GED 2248, must be taken by the students of the CSE department.

Course Code	UNESCO Code	Course Title	Credit
GED Tier I	Course-wise (varies)	Arts and Humanities	3
GED Tier II SSC 2243	0521-000-2243	Environment and Sustainability (for the students of the Engineering School)	3
GED Tier III NSC 2248	0417-000-2248	Industrial Management	3

Course Catalogue

Mathematics and Statistics Courses (4 courses /12 credits)

Course Code	UNESCO Code	Course Title	Credit	Prerequisite
MAT 1103	0541-014-1103	Calculus and Differential Equations	3	
MAT1203	0541-014-1203	Coordinate Geometry, Vector Analysis and Complex Variables	3	MAT 1103
MAT2103	0541-014-2103	Linear Algebra and Numerical Analysis	3	MAT 1203
STA 2101	0542-014-2101	Probability and Statistics	3	

Basic Science Courses (2 Theory + 1 Lab /7 credits)

Course Code	UNESCO Code	Course Title	Credit	Prerequisite
PHY1101	0533-014-1101	Physics I	3	
PHY1102	0533-014-1102	Physics I Lab	1	
PHY1301	0533-014-1301	Physics II	3	PHY1101

Other Engineering Courses (2 Theory + 2 Labs / 8 credits)

Course Code	UNESCO Code	Course Title	Credit	Prerequisite
EEE 1101	0713-016-1101	Electrical Circuit 1	3	
EEE1102	0713-016-1102	Electrical Circuit 1 Lab	1	
EEE1301	0714-016-1301	Electronic Device and Circuits 1	3	EEE 1101, EEE 1102
EEE1302	0714-016-1302	Electronic Device and Circuits 1 Lab	1	EEE 1101, EEE 1102

Major Core Courses (16 Theory + 14 Lab + Capstone Project + Internship / 67 credits)

Course Code	UNESCO Code	Course Title	Credit	Prerequisite
CSE1201	0613-014-1201	Structured Programming	3	
CSE1202	0613-014-1202	Structured Programming LAB	1	
CSE1203	0613-014-1203	Discrete Mathematics	3	
CSE1301	0613-014-1301	Data Structures	3	CSE1201 & CSE1202
CSE1302	0613-014-1302	Data Structures LAB	1	CSE1201 & CSE1202
CSE2101	0613-014-2101	Digital Logic Design	3	
CSE2102	0613-014-2102	Digital Logic Design Lab	1	
CSE2103	0613-014-2103	Object Oriented Programming	3	CSE1301 & CSE1302
CSE2104	0613-014-2104	Object Oriented Programming LAB	1	CSE1301 & CSE1302
CSE2201	0613-014-2201	Algorithms	3	CSE1301, CSE1302, CSE1203
CSE2202	0613-014-2202	Algorithms LAB	1	CSE1301, CSE1302, CSE1203
CSE2203	0613-014-2203	Computer Organization and Architecture	3	CSE2101, CSE2102
CSE2200	0613-014-2200	Design Project-I	1	CSE2103, CSE2104
CSE2301	0612-014-2301	Database Management System	3	
CSE2302	0612-014-2302	Database Management System Lab	1	

CSE2303	0613-014-2303	Automata and Theory of Computation	3	CSE2201, CSE2202
CSE2305	0613-014-2305	Operating Systems	3	CSE2203
CSE2306	0613-014-2306	Operating Systems Lab	1	CSE2203
CSE3101	0613-014-3101	Microprocessor and Microcontroller	3	CSE2203
CSE3102	0613-014-3102	Microprocessor and Microcontroller Lab	1	CSE2203
CSE3103	0613-014-3103	System Analysis and Design	3	CSE2103, CSE2104, CSE 2200
CSE3104	0613-014-3104	System Analysis and Design Lab	1	CSE2103, CSE2104, CSE 2200
CSE3120	0613-014-3120	Web Programming	1	CSE2301 & CSE2302
CSE3201	0619-014-3201	Artificial Intelligence & Machine Learning	3	CSE 2201, CSE2202, STA2101, MAT2103
CSE3202	0619-014-3202	Artificial Intelligence & Machine Learning Lab	1	CSE 2201, CSE2202, STA2101, MAT2103
CSE3203	0613-014-3203	Software Engineering	3	CSE3103, CSE3104
CSE3200	0613-014-3200	Design Project-II	1	CSE2301, CSE2302, CSE3103, CSE3104, CSE2200
CSE3205	0612-014-3205	Computer Networks	3	
CSE3206	0612-014-3206	Computer Networks Lab	1	
CSE3301	0612-014-3301	Cyber Security	3	CSE 2305, CSE 2306, CSE 3205, CSE 3206
CSE4098A	0613-014-4098A	Capstone Project 1	1	Completion of 105 credits, CSE3103, CSE3203, CSE2200 & CSE3200
CSE4098B	0613-014-4098B	Capstone Project 2	1	CSE 4098-A

CSE4098C	0613-014-4098C	Capstone Project 3	2	CSE 4098-B
CSE4099A/ CSE4099B	0613-014-4099A/0613-014-4099B	Internship / Thesis	1	CSE 4098-C CSE 4098-C

Concentration Group (4 theory + 1 lab /13 credits)

The students will choose a concentration group consisting of four elective courses. These groups are designed to provide undergraduate specialization in specific topics. The offered groups are:

- I. Computational Theory
- II. Network and Communications
- III. Security
- IV. Systems
- V. Data Science
- VI. Software Engineering
- VII. Hardware
- VIII. Information and Communication Technology

Students will have to take four elective theory courses and one lab course (associated with any of those four courses) from one of the concentration groups.

Computational Theory Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4401	0613-014-4101	Computer Graphics	3
CSE 4402	0613-014-4402	Computer Graphics Lab	1
CSE 4403	0613-014-4403	Advanced Algorithm	3
CSE 4405	0613-014-4205	Compiler Design	3
CSE 4406	0613-014-4206	Compiler Design Lab	1
CSE 4407	0613-014-4207	Basic Graph Theory	3
CSE 4409	0613-014-4209	Mathematical Analysis for Computer Science	3
CSE 4411	0613-014-4211	Computational Geometry	3
CSE 4413		Topics of Current Interest in Computational Theory	3

Network and Communications Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4415	0612-014-4115	Data Communication	3
CSE 4416	0612-014-4116	Data Communication Lab	1
CSE 4417	0613-014-4417	Internet of Things	3
CSE 4418	0613-014-4418	Internet of Things Lab	1
CSE 4419	0612-014-4219	Network Security	3

CSE 4420	0612-014-4220	Network Security Lab	1
CSE 4421	0612-014-4221	Wireless and Cellular Communication	3
CSE 4423	0688-014-4223	Digital Signal Processing	3
CSE 4425	0612-014-4225	Advanced Network Services and Management	3
CSE 4427		Topics of Current Interest in Network and Communications	3

Security Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4419	0612-014-4219	Network Security	3
CSE 4420	0612-014-4220	Network Security Lab	1
CSE 4429	0613-014-4129	Software Security	3
CSE 4430	0613-014-4130	Software Security Lab	1
CSE 4431	0688-014-4131	Blockchain	3
CSE 4433	0612-014-4233	Cryptography	3
CSE 4435	0688-014-4235	ICT Law, Policy and Ethics	3
CSE 4437	0688-014-4237	Digital Forensics and Incident Response	3
CSE 4439		Topics of Current Interest in Security	3

Computer Systems Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4441	0613-014-4141	Real-time Embedded Systems	3
CSE 4442	0613-014-4142	Real-time Embedded Systems Lab	1
CSE 4443	0612-014-4143	Distributed Systems	3
CSE 4445	0613-014-4145	Simulation and Modeling	3
CSE 4446	0613-014-4146	Simulation and Modeling Lab	1
CSE 4447	0688-014-4247	Introduction to Robotics	3
CSE 4449	0612-014-4249	Cloud Computing	3
CSE 4451	0612-014-4251	Advanced Database Management Systems	3
CSE 4453		Topics of Current Interest in Computer Systems	3

Data Science Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4455	0619-014-3355	Data Mining	3
CSE 4457	0619-014-4157	Data Science	3
CSE 4458	0619-014-4158	Data Science Lab	1
CSE 4459	0619-014-4159	Big Data Analytics	3
CSE 4460	0619-014-4160	Big Data Analytics Lab	1
CSE 4461	0613-014-4261	Digital Image Processing	3
CSE 4462	0613-014-4262	Digital Image Processing Lab	1
CSE 4463	0688-014-4263	Introduction to Bioinformatics	3
CSE 4465	0619-014-4265	Natural Language Processing	3
CSE 4467		Topics of Current Interest in Data Science	3

Software Engineering Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4469	0613-014-4169	Software Requirements Specification and Analysis	3
CSE 4471	0613-014-4171	Design Patterns	3
CSE 4473	0613-014-4173	Software Testing and Quality Assurance	3
CSE 4474	0613-014-4174	Software Testing and Quality Assurance Lab	1
CSE 4475	0613-014-4275	Mobile Application Development	3
CSE 4477	0613-014-4277	Advanced Programming	3
CSE 4478	0613-014-4278	Advanced Programming Lab	1
CSE 4479	0613-014-4279	Human Computer Interaction	3
CSE 4481		Topics of Current Interest in Software Engineering	3

Information and Communication Technology Group

Course Code	UNESCO Code	Course Title	Credits
CSE 4483	0688-014-4183	Enterprise Systems: Concepts and Practice	3
CSE 4484	0688-014-4184	Enterprise Systems: Concepts and Practice Lab	1
CSE 4485	0688-014-4185	Electronic Business	3
CSE 4473	0613-014-4173	Software Testing and Quality Assurance	3
CSE 4474	0613-014-4174	Software Testing and Quality Assurance Lab	1
CSE 4475	0613-014-4275	Mobile Application Development	3

CSE4487	0613-014-4287	UI: Concepts and Design	3
CSE4489	0688-014-4289	IT Audit: Concepts and Practice	3
CSE4449	0612-014-4249	Cloud Computing	3
CSE4491	0688-014-4291	ICT for Development	3
CSE4493		Topics of Current Interest in Information and Communication Technology	3

Minor/Optional

For CSE Students

CSE students, choosing the Optional, have to complete three courses offered by any department other than CSE as an optional courses.

Students choosing a Minor must take a total of 5 courses from any of the undergraduate programs. The courses must be chosen from that program's specified courses for Minor.

Minor in CSE (For Students of other departments)

Students of other departments wishing to do a Minor in CSE can choose from courses listed in the table below.

Course Code	UNESCO Code	Course Title	Credits
CSE 1201	0613-014-1201	Structured Programming	3
CSE1202	0613-014-1202	Structured Programming Lab	1
CSE2103	0613-014-2103	Object Oriented Programming	3
CSE2104	0613-014-2104	Object Oriented Programming Lab	1
CSE1301	0613-014-1301	Data Structure	3
CSE2301	0612-014-2301	Database Management System	3
CSE2203	0613-014-2203	Computer Organization and Architecture	3
CSE3201	0619-014-3201	Artificial Intelligence & Machine Learning	3
CSE4417	0613-014-4417	Internet of Things	3
CSE4465	0619-014-4265	Natural Language Processing	3

*Students who want to take CSE 1301 or CSE 2103 must take the corresponding Lab courses.

Capstone Project

The Department of CSE requires each graduating student of BSc in CSE program to complete a three-term long capstone project (4 credits). The term-wise credit distribution will be 1+1+2 = 4. To enroll in the Capstone Project, students must complete at least 105 credits including all the major core courses, math and science courses, Ethics, Language and Communication-related courses. Students must register for the Capstone Project in Term 9 and complete the project in Term 11. Students will have to submit a project report and present their project work before a panel of examiners by the end of Term 11.

Grading will be done based on the report, presentation, continuous assessment, and other assessment methods as decided by the department through 3 semesters. This project is intended to provide a culminating experience that allows a student to demonstrate proficiency in several of the learning outcomes that are stated by his or her degree program. In the Capstone Project, students will apply and cumulate the knowledge

and skills that they gathered through theory and lab courses in their program. For this reason, each approved Capstone Project should include and properly address the following components:

- Problem Analysis
- Design and Development of Solutions
- Investigation
- Project Management
- Teamwork
- Requirements Analysis
- Design and Implementation
- Ethics
- Societal and Environmental Impacts
- Communication
- Lifelong Learning
- Modern Tools Usage and Contemporary Issues

Sample 4-Year Distribution of Courses

Term	Course Code	UNESCO Code	Course Title	Credits
Term 1	MAT1103	0541-014-1103	Calculus and Differential Equations	3
	GEF1101/ ELL0099	0231-000-1101/ 0231-000-0099	Academic English I/ Remedial English	3
	PHY1101	0533-014-1101	Physics I	3
	PHY1102	0533-014-1102	Physics Lab	1
	ESK1110	0031-000-1110	Study Skills	
Term 2	MAT 1203	0541-014-1203	Coordinate Geometry, Vector Analysis and Complex Variables	3
	CSE1201	0613-014-1201	Structured Programming	3
	CSE1202	0613-014-1202	Structured Programming Lab	1
	CSE1203	0613-014-1203	Discrete Mathematics	3
	UCC1101	0232-000-1101	Bangla Bhasha O Sahitya	3
Term 3	ESK1111	0031-000-1111	Healthy Life Skills	
	PHY1301	0533-014-1301	Physics II	3
	CSE1301	0613-014-1301	Data Structures	3
	CSE1302	0613-014-1302	Data Structures Lab	1
	EEE1101	0713-016-1101	Electrical Circuit 1	3
	EEE1102	0713-016-1102	Electrical Circuit 1 Lab	1
	UCC1201	0222-000-1201	History of the Emergence of Independent Bangladesh	3
Term 4	ESK1112	0031-000-1112	Social Skills	

Term 4	MAT2101	Delete row	This row will be deleted	3
	MAT 2103	0541-014-2103	Linear Algebra and Numerical Analysis	3
	GEF 1201	0231-000-1201	Academic English II	3
	CSE 2101	0613-014-2101	Digital Logic Design	3
	CSE 2102	0613-014-2102	Digital Logic Design Lab	1
	CSE2103	0613-014-2103	Object Oriented Programming	3
	CSE2104	0613-014-2104	Object Oriented Programming Lab	1
	ESK1113	0031-000-1113	Professional Skills	
Term 5	CSE2201	0613-014-2201	Algorithm	3
	CSE2202	0613-014-2202	Algorithm Lab	1
	STA2101	0542-014-2101	Probability and Statistics	3
	EEE1301	0714-016-1301	Electronic Device and Circuits I	3
	EEE1302	0714-016-1302	Electronic Device and Circuits 1 Lab	1
	CSE2200	0613-014-2200	Design Project-I	1
Term 6	CSE2301	0612-014-2301	Database Management System	3
	CSE2302	0612-014-2302	Database Management System Lab	1
	CSE2303	0613-014-2303	Automata and Theory of Computation	3
	CSE2305	0613-014-2305	Operating Systems	3
	CSE2306	0613-014-2306	Operating Systems Lab	1
	GED2159	0223-000-2159	Professional Ethics (GED 5)	3
Term 7	CSE 3101	0613-014-3101	Microprocessor and Microcontroller	3
	CSE 3102	0613-014-3102	Microprocessor and Microcontroller Lab	1
	CSE3103	0613-014-3103	System Analysis and Design	3
	CSE3104	0613-014-3104	System Analysis and Design Lab	1
	CSE3120	0613-014-3120	Web Programming	1
	GED Elective-1(HUM)		One Course from Tier I (Arts and Humanities)	3

Term 8	CSE3201	0619-014-3201	Artificial Intelligence & Machine Learning	3
	CSE3202	0619-014-3202	Artificial Intelligence & Machine Learning Lab	1
	CSE3203	0613-014-3203	Software Engineering	3
	CSE3200	0613-014-3200	Design Project-II	1
	CSE3205	0612-014-3205	Computer Networks	3
	CSE3206	0612-014-3206	Computer Networks Lab	1
	SSC2243(GED Elective 2)	0521-000-2243	Environment and Sustainability (Social Science)	3
Term 9	Major Elective-1			3
	CSE3301	0612-014-3301	Cyber Security	3
	Optional/ Minor-1			3
	CSE4098A	0613-014-4098A	Capstone Project-1	1
	NSC 2248 (GED Elective-3)	0417-000-2248	Industrial Management (Natural Science)	3
Term 10	Major Elective-2+Lab			
	Major Elective 3			
	Optional /Minor 2			
	CSE 4098B	0613-014-4098B	Capstone Project-2	1
Term 11	Major Elective 4			
	Optional/ Minor-3			
	CSE4098C	0613-014-4098C	Capstone Project-3	2
Term 12	CSE4099-A	0613-014-4099A	Internship	1
	CSE4099-B	0613-014-4099B	Thesis	1

***NOTE:** All courses may not be available every term. The Computer Science and Engineering Department reserves the right to add, drop or substitute individual courses, subject to review from academic authorities.